HOW IT WORKS

How a print job runs a language model

TONER-1 is the most capable language model ever to run entirely inside a print job.* It is one 595 KB PostScript file. Send it to a laser printer and the printer's own processor writes a short story, then prints it. The computer that sent it does no math.

01

A print job is a program

PostScript, the language most laser printers speak, is not a file format but a complete programming language, stack-based like Forth. Send `3 4 add` and the interpreter leaves 7 on its stack. It has variables, loops and floating-point math. A print job is simply a program; the printed page is a side effect of the drawing commands it calls.

02

Printers are computers

To run those programs, every PostScript printer carries its own processor and memory. The 1985 Apple LaserWriter had a 12 MHz Motorola 68000 and 1.5 MB of RAM, a faster CPU than the original Macintosh it plugged into (a 68000 at about 7.8 MHz). Modern printers have far more, and spend most of it waiting for spreadsheets.

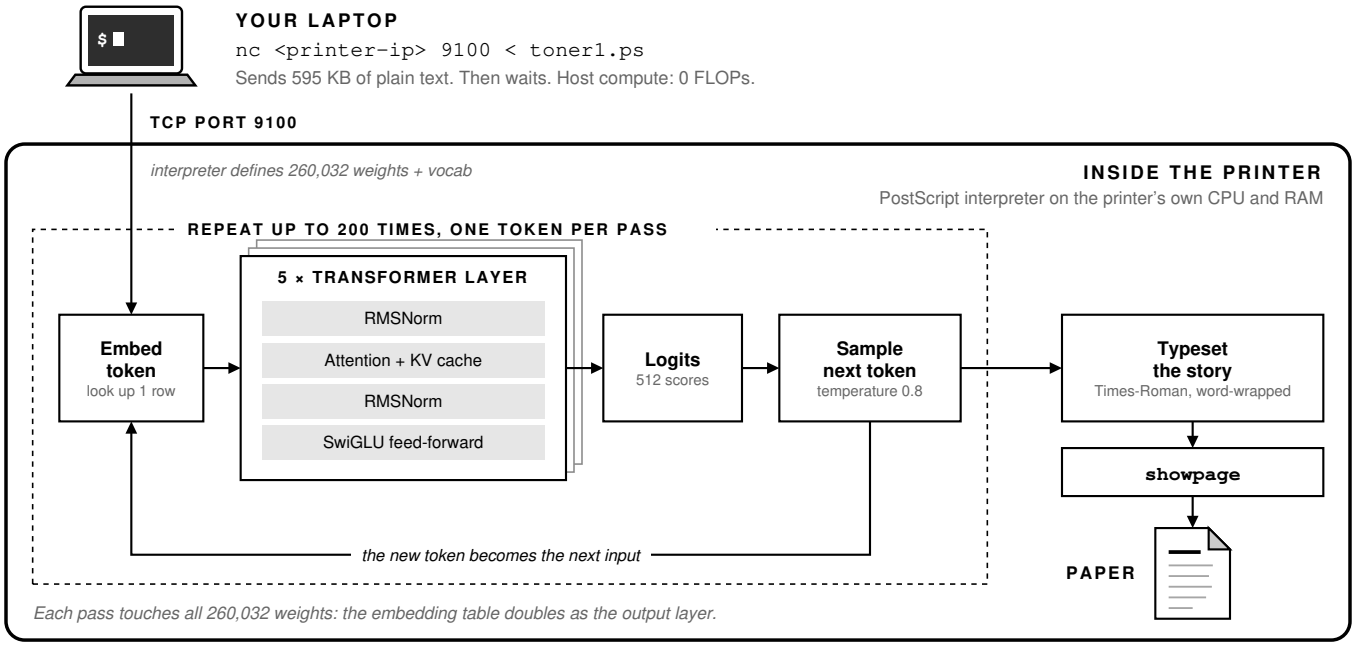
03

An LLM is just arithmetic

Strip away the mystique and a language model is a long list of multiplications and additions: matrix-vector products, a normalization step, a softmax that turns scores into probabilities, and a weighted coin flip that picks the next word piece. Any language with loops and multiplication can run one. PostScript has both.

THE PIPELINE

What happens after you send toner1.ps to port 9100



260,032 parameters
int8, one scale per row

595 KB the entire print job
weights + vocab + program

≈55 tok/s in Ghostscript 10.07
Apple-silicon Mac, measured

0.3–5 tok/s on a real printer
estimated, not yet measured

0 FLOPs on the host
the laptop only sends bytes

** Among int8, 260,032-parameter, English-language, children's-story models executed by a PostScript interpreter, as of this printing. Ghostscript counts as a printer for the purposes of this footnote. Model: Karpathy's llama2.c stories260K, trained on TinyStories. Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.*

INSIDE THE FILE

What's inside the 595 KB print job

BYTES IN TONER1.PS BY PURPOSE: 595,177 TOTAL, 454 LINES OF PLAIN TEXT



THE DATA, LINES 1 TO 106

```
 1 %!PS-Adobe-3.0
 7 /D 64 def /HID 172 def /NL 5 def /NH 8 def /NKV 4 def
     /V 512 def /SEQ 512 def /HS 8 def /KVD 32 def
     /KVMUL 2 def /NPARAMS 260032 def
```

```
16 /WQ [
17 [[0.00241668 0.00436529 0.00353898 0.00237025 ... 0.0012341]
18 [<92a17c888b708f7c23077585b2b3418c5c1e784b...> <b39d...> ...]]
```

```
row 0 bytes        92       a1       7c       88
minus 128          18       33       -4        8
× 0.00241668  0.0435   0.0798  -0.0097   0.0193
```

1. **The model card is one line.** Line 7 defines width 64, 5 layers, 8 query heads sharing 4 key/value heads and a 512-token vocabulary. Nothing runs yet; names are just defined. Line 1 makes it a print job; lines 2 to 6 are comments for the spooler.

2. **Weights are bytes.** For each matrix row, export.py takes the largest absolute weight m and stores scale = m/127. Each weight is rounded to an integer from -127 to 127, plus 128, and written as one byte of a hex string. Two hex characters per weight: that is most of the file.

THE PROGRAM, LINES 107 TO 454

```
128 % o x M mv  ;  M = [scales rows], rows ... offset by 128
129 /mv { /mM exch def /mx exch def /mo exch def
130   /mS mM 0 get def /mQ mM 1 get def /mn mx length 1 sub def
131   /msx 0 mx { add } forall 128 mul def
132   0 1 mQ length 1 sub { /mi exch def /mrow mQ mi get def
133     0 0 1 mn { dup mrow exch get exch mx exch get mul add } for
134     msx sub mS mi get mul mo mi 3 -1 roll put } for
135 } bind def
```

3. **mv does the heavy lifting.** Matrix times vector: about 259,000 multiply-adds per token run through these eight lines. `get` on a string returns a byte as an integer. Rather than subtract 128 from every byte, mv takes the raw dot product, subtracts `msx` (128 times the sum of the input), then applies the row's scale.

```
138 /rope { /rn exch def /rv exch def
139   0 2 rn 1 sub { /ri exch def
140     /hd ri HS mod def
141     pos 10000 hd HS div exp div 57.29577951 mul /ang exch def
142     /fcr ang cos def /fci ang sin def
```

4. **RoPE, in degrees.** Rotary embeddings turn query/key pairs by `pos / 10000^(hd/8)` radians, but PostScript's `sin` and `cos` take degrees. Hence 57.29577951 (180/pi). Without it, every position would get the wrong rotation.

```
117 SEED srand 4 { rand pop } repeat
203 /r rand 2147483647 div ssum mul def /cum 0 def
      /pick V 1 sub def
204 0 1 V 1 sub { /j exch def /cum cum logits j get add def
      cum r gt { /pick j def exit } if } for
```

5. **Sampling with rand.** The logits become a softmax at temperature 0.8. PostScript's built-in `rand` picks a point along the total, and the loop walks the running sum until it passes it. Line 117 skips four tiny first draws. Seed 42: same interpreter, same story.

```
256 % ... typesetting: the same print job lays out the page ...
367 fs /Times-Roman F ...
448 9 /Helvetica-Bold F 72 79 moveto (Host compute: ...) show
452 showpage
```

6. **Then, and only then, toner.** After 200 tokens (or an end-of-story token), the same job word-wraps the story in Times-Roman, draws the run card and calls `showpage`. Until then, the printer was just a computer.

PRINT IT YOURSELF — BRING YOUR OWN PRINTER

```
nc <printer-ip> 9100 < toner1.ps
```

TONER-1 Pro: toner1_15M.ps, 15M parameters, 31.6 MB. Same command. Much slower.

- **Needs real PostScript.** A PostScript interpreter, Level 2 or later, is required. PCL- or GDI-only printers may print the file as raw text.
- **Mind the limits.** Long jobs can hit the job timeout or the VM limit. Ghostscript uses about 3.5 MB of VM.
- **Expect minutes.** An estimated 0.3 to 5 tok/s is roughly 1 to 10 minutes per 200-token page. It is thinking.
- **Ask first.** Only use printers you own or have permission to use. The queue behind you has feelings.

STATUS ☑ Verified in Ghostscript 10.07: 200 tokens, about 55 tok/s, about 3.5 MB of VM ☐ **Not yet run on a physical printer**

Excerpts are verbatim from toner1.ps; gray . . . marks trims, and long lines are re-wrapped or dedented. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in Ghostscript only; PSChess played chess on real printers. TONER-1 is built to run on an actual printer; that run is pending. Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.

TONER-1: a frontier model with a paper trail.

State of the art among language models that run entirely inside a print job.

> The most capable open-weight, US-Letter-native, sub-300K-parameter, PostScript-native, zero-host-FLOP, duplex-ready language model running on hardware already in your copy room. *Evaluated on 20 lb bond at room temperature.*

SPECIFICATIONS

Base model	Karpathy's llama2.c stories260K, trained on TinyStories
Parameters	260,032
Architecture	Llama-2 decoder: 5 layers, dim 64, 8 query heads / 4 KV heads, head size 8, SwiGLU FFN (hidden 172), RoPE, RMSNorm
Vocabulary	512-token SentencePiece
Quantization	int8 weights, one scale per row; activations in PostScript reals; embedding shared with the output head
Context window	8.5 × 11 in. (512 tokens)
KV cache	Persistent, archival, recyclable
Job size	595 KB, one file: `toner1.ps`. Weights, vocabulary, a forward pass of about 100 lines and its own page layout
Speed, measured	About 55 tok/s in Ghostscript 10.07 on an Apple-silicon Mac, in about 3.5 MB of PostScript VM
Speed, printer	0.3 to 5 tok/s, estimated, not yet measured. A 200-token page takes roughly 1 to 10 minutes
Time to first token	Includes fuser warm-up
Accelerator	The office's hottest inference chip (fuser, typically about 200 °C)
Host compute	0 FLOPs. The host only sends bytes: `nc <printer-ip> 9100 < toner1.ps`
Output modality	Native toner
Open weights	Literally. Hex appendix available; OCR to load
Pricing	Inference cost includes toner

INTENDED USE

Writing short children's stories, one US Letter sheet at a time, on a printer you own or have permission to use. Demonstrating that a print job is a program. *Out of scope: prompts. The model chooses the plot.*

LIMITATIONS

- It is 260K parameters.
- It mostly writes about Lily and Tim.
- It will hallucinate a quiet salad.
- No prompt input: every story starts from the beginning-of-sequence token.
- Needs a PostScript interpreter, Level 2 or later. PCL- or GDI-only printers may print the file as text.
- Long jobs may hit the printer's job timeout or run out of VM.

SAFETY

- Refuses to generate while Tray 2 is open. Close the tray to proceed.
- **Air-gapped mode:** unplug Ethernet once the job has uploaded.
- **Kill switch:** the power button.
- Outputs are fused to paper and can no longer execute.

PRODUCT LINEUP

TONER-1 — AVAILABLE

STORIES260K / 595 KB JOB

The everyday copy-room model. Under 4 s per page in Ghostscript; an estimated 1 to 10 minutes on a real printer.

TONER-1 Pro — EST. OVERNIGHT

STORIES15M / 31.6 MB JOB

Same print job, about 58 times the parameters. Estimated: send it at close of business, collect your story in the morning.

TONER-1 Classic — PLANNED

1985 LASERWRITER / 68000

A 1985-native AI PC. The LaserWriter's 12 MHz 68000 was faster than the 7.8 MHz Macintosh it was plugged into.

TONER-1 is a working print job and a parody of frontier-model launches. This handout is a PostScript program, too.
Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.

Undefeated inside a print job.

TONER-1 posts a result on every benchmark it can enter. Frontier models from major labs were offered identical conditions: one print job, one sheet, zero host FLOPs.

FIGURE 1 HEADLINE RESULTS (PARODY)

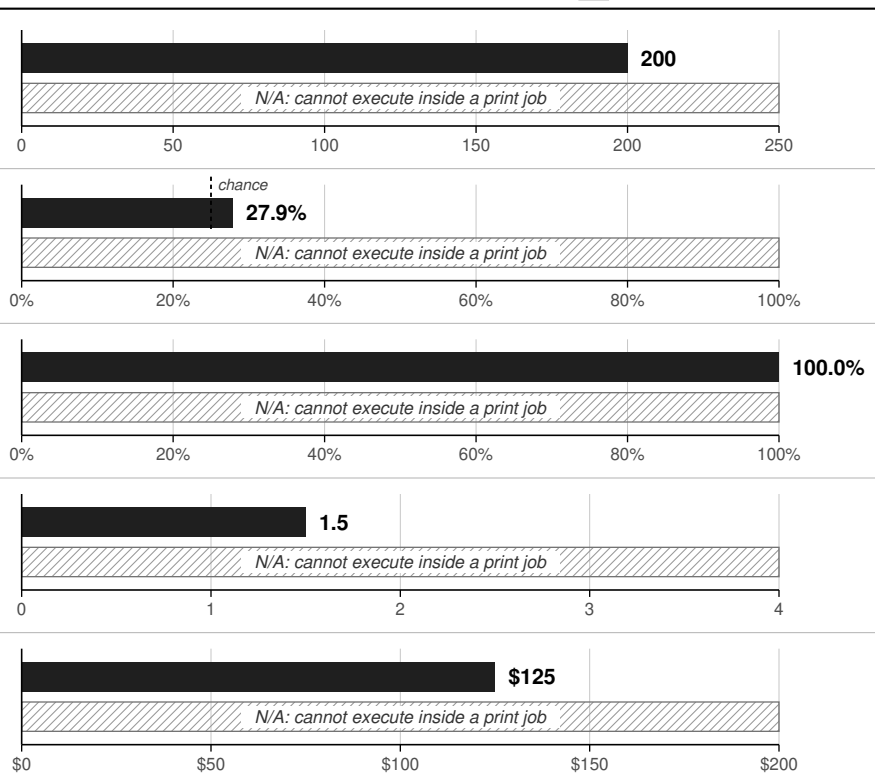
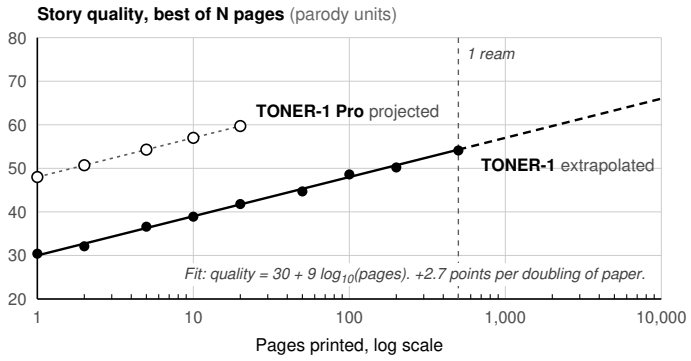


FIGURE 2 TEST-TIME SCALING



EVALUATION SETUP

Interpreter	Ghostscript 10.07, Apple-silicon Mac
Printer	Office laser, PostScript emulation (pending)
Speed	Measured in Ghostscript; printer estimated
Paper	20 lb bond, US Letter, simplex
Sampling	Temperature 0.8, seed 42, 200 steps
Prompt	None. Every run starts from BOS.
Ambient	Room temp.; fuser typically about 200 °C
Frontier models	Not run: no print-job support

REPRODUCE

```
nc <printer-ip> 9100 < toner1.ps
```

Parody benchmarks. Not real evaluations. TONER-1 speed measured in Ghostscript; printer speed estimated. Frontier models from major labs were not run and are not named; N/A means only that they cannot execute inside a print job. Figure 2 is parody data. TONER-1 is Karpathy's llama2.c stories260K, exported to int8 and run by a PostScript forward pass. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in PostScript under Ghostscript, and PSChess played chess on real printers. TONER-1 is built to run on an actual printer; that run is pending. Only use printers you own or have permission to use.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play outside and seek. One day, she went to the park with her mommy. They saw a cute dog and a small dog named Max who was playing with a ball. Lily was very excited and wanted to help her mommy.

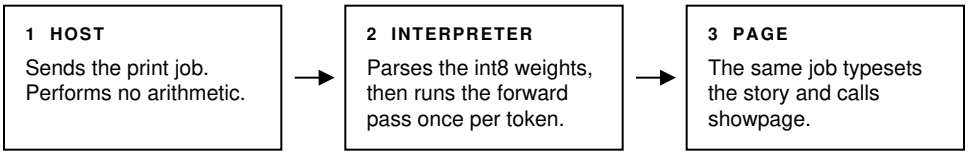
Lily opened the dog and saw a big bell with a ball. She asked her mommy, “Can I play with you?” Her mommy said, “Okay, Max.” Lily was happy to hear that and said, “Thank you, Max. I am you so restless.”

TOKEN STREAM

All 184 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

`·Once` `·upon` `·a` `·time` `,` `·there` `·was` `·a` `·little` `·g` `ir` `l` `·named` `·Lily` `.` `·She`
`·lo` `ved` `·to` `·play` `·` `out` `s` `id` `e` `·and` `·s` `e` `e` `k` `.` `·One` `·day` `,` `·she` `·w` `ent`
`·to` `·the` `·p` `ar` `k` `·with` `·her` `·mom` `my` `.` `·They` `·saw` `·a` `·c` `ut` `e` `·do` `g` `·and`
`·a` `·s` `m` `all` `·do` `g` `·named` `·M` `a` `x` `·w` `h` `o` `·was` `·play` `ing` `·with` `·a` `·b` `all` `.`
`·Lily` `·was` `·very` `·e` `x` `c` `it` `ed` `·and` `·want` `ed` `·to` `·he` `l` `p` `·her` `·mom` `my` `.` `¶`
`L` `ily` `·o` `p` `en` `ed` `·the` `·do` `g` `·and` `·saw` `·a` `·big` `·be` `ll` `·with` `·a` `·b` `all` `.`
`·She` `·a` `s` `ked` `·her` `·mom` `my` `,` `·"` `C` `an` `·I` `·play` `·with` `·you` `?` `"` `·H` `er` `·mom`
`my` `·said` `,` `·"` `O` `k` `ay` `,` `·M` `a` `x` `.` `"` `·Lily` `·was` `·happy` `·to` `·he` `ar` `·that`
`·and` `·said` `,` `·"` `T` `h` `an` `k` `·you` `,` `·M` `a` `x` `.` `·I` `·a` `m` `·you` `·so` `·` `re` `st` `le` `s`
`s` `.` `"`

HOW THIS PAGE WAS MADE



TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
184	3.5	52.3	20	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	End-of-story token, after 184 tokens

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play outside and pick pieces. One day, she went for a walk in the pond. She saw a yellow meal that she lost her toy car.

Lily went to the meadow, but her mom asked her to help her. Lily told her about her toy car's toys and her king. The mechanic found the mum and started to cry. Suddenly, a boy saw a big tree and said, "Hello, I am sorry. I am a cheese. I lost me."

Lily and…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

`·Once` `·upon` `·a` `·time` `,` `·there` `·was` `·a` `·little` `·g` `ir` `l` `·named` `·Lily` `.` `·She`
`·lo` `ved` `·to` `·play` `·` `out` `s` `id` `e` `·and` `·p` `i` `ck` `·p` `i` `e` `ce` `s` `.` `·One` `·day` `,`
`·she` `·w` `ent` `·for` `·a` `·wa` `l` `k` `·in` `·the` `·p` `o` `nd` `.` `·She` `·saw` `·a` `·y` `e` `ll` `ow`
`·m` `e` `a` `l` `·that` `·she` `·lo` `st` `·her` `·to` `y` `·c` `ar` `.` `¶` `L` `ily` `·w` `ent` `·to` `·the` `·m`
`e` `ad` `ow` `,` `·but` `·her` `·mom` `·a` `s` `ked` `·her` `·to` `·he` `l` `p` `·her` `.` `·Lily` `·to` `ld`
`·her` `·a` `b` `out` `·her` `·to` `y` `·c` `ar` `'` `s` `·to` `y` `s` `·and` `·her` `·k` `ing` `.` `·The` `·m` `e`
`ch` `an` `i` `c` `·f` `ou` `nd` `·the` `·m` `u` `m` `·and` `·st` `ar` `t` `ed` `·to` `·c` `r` `y` `.` `·S` `u` `d` `d`
`en` `l` `y` `,` `·a` `·b` `o` `y` `·saw` `·a` `·big` `·t` `re` `e` `·and` `·said` `,` `·"` `H` `e` `ll` `o` `,` `·I`
`·a` `m` `·s` `or` `r` `y` `.` `·I` `·a` `m` `·a` `·c` `he` `e` `se` `.` `·I` `·lo` `st` `·m` `e` `.` `"` `·` `¶` `L` `ily`
`·and`

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	3.9	51.7	40	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play outside in the park with her friends. One day, they found a big, round bird on the ground. Lily asked her friend, Sarah, “What’s wrong, I will make it breakfast,” her friend said.

Lily’s friend said, “I want to play with a silly bird!” Sarah gave Sarah a bird’s bag and took a big bite. Sarah was very happy and played with her toys all day long. When they reached, Lily felt better!

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

`·Once` `·upon` `·a` `·time` `,` `·there` `·was` `·a` `·little` `·g` `ir` `l` `·named` `·Lily` `.` `·She`
`·lo` `ved` `·to` `·play` `·` `out` `s` `id` `e` `·in` `·the` `·p` `ar` `k` `·with` `·her` `·friend` `s` `.`
`·One` `·day` `,` `·they` `·f` `ou` `nd` `·a` `·big` `,` `·r` `ou` `nd` `·b` `ir` `d` `·on` `·the` `·g` `r` `ou`
`nd` `.` `·Lily` `·a` `s` `ked` `·her` `·friend` `,` `·S` `ar` `a` `h` `,` `·"` `W` `h` `at` `’` `s` `·w` `r` `on` `g`
`,` `·I` `·w` `il` `l` `·m` `a` `ke` `·it` `·b` `re` `a` `k` `f` `a` `st` `,` `"` `·her` `·friend` `·said` `.` `¶` `L`
`ily` `’` `s` `·friend` `·said` `,` `·"` `I` `·want` `·to` `·play` `·with` `·a` `·s` `il` `l` `y` `·b` `ir` `d`
`!` `"` `·S` `ar` `a` `h` `·g` `a` `ve` `·S` `ar` `a` `h` `·a` `·b` `ir` `d` `’` `s` `·b` `a` `g` `·and` `·to` `o` `k` `·a`
`·big` `·b` `it` `e` `.` `·S` `ar` `a` `h` `·was` `·very` `·happy` `·and` `·play` `ed` `·with` `·her` `·to` `y`
`s` `·a` `ll` `·day` `·l` `on` `g` `.` `·` `W` `he` `n` `·they` `·` `re` `a` `c` `he` `d` `,` `·Lily` `·f` `e` `l` `t`
`·b` `et` `t` `er` `!`

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	3.8	53.2	22	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200, reached before the end-of-story token.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play outside in the park. One day, she saw a big, red ball. She wanted to play with it, but it was too high. She asked her mom, “Can I play with the ball?” Her mom said, “Yes, it’s time to go home.”

Lily was very excited to see the ball. She put the ball in the ball and put it in the ball. But the ball was too high. Lily was sad. She did not want to share her ball. She said, “I will help you find it.”

Lily was sad and didn’t k…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.



HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	**3.6**	**54.9**	**42** (unused)	**0** greedy

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

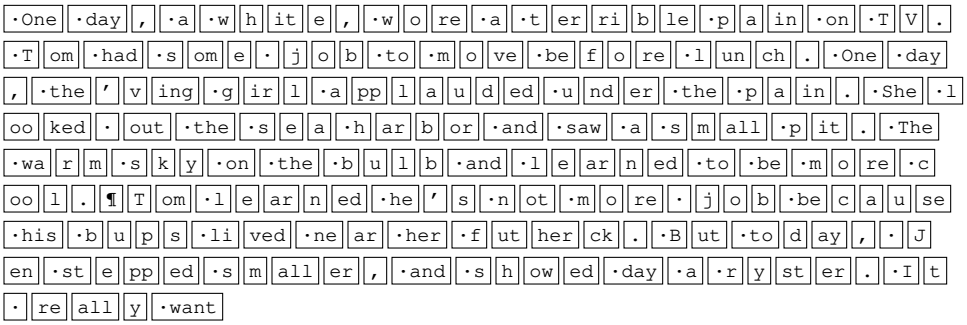
Unedited model output. No prompt: generation starts at the BOS token.

One day, a white, wore a terrible pain on TV. Tom had some job to move before lunch. One day, the'ving girl applauded under the pain. She looked out the sea harbor and saw a small pit. The warm sky on the bulb and learned to be more cool.

Tom learned he's not more job because his bups lived near her futherck. But today, Jen stepped smaller, and showed day a ryster. It really want…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.



HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	3.7	54.1	6	1.4

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.

TONER-1 Pro

15,191,712 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a big hall. In the hall, there were many people who played and danced. They were so happy! One day, a little girl named Lily went to the hall to play. She was excited to be independent and play by herself.

Lily saw a big dog near the room. The dog was not nice. He barked at Lily and bit her. Lily felt sad and scared. She started to cry.

A kind man saw Lily crying and came to help. He talked to the dog and said, “Please be kind to Lily.” The dog stopped barking and did not bite Lily. Instead, the dog played with Lily and they had fun together. Lily was happy and learned that being independent is good.

TOKEN STREAM

The first 142 of 168 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once	·upon	·a	·time	,	·there	·was	·a	·big	·hall	.	·In	·the	·hall	,	·there		
·were	·many	·people	·who	·played	·and	·dan	ced	.	·They	·were	·so	·happy	!				
·One	·day	,	·a	·little	·girl	·named	·L	ily	·went	·to	·the	·hall	·to	·play			
.	·She	·was	·excited	·to	·be	·independent	·and	·play	·by	·herself	.	¶	L				
ily	·saw	·a	·big	·dog	·near	·the	·room	.	·The	·dog	·was	·not	·nice	.	·He		
·b	ark	ed	·at	·L	ily	·and	·bit	·her	.	·L	ily	·felt	·sad	·and	·sc	ared	.
·She	·started	·to	·cry	.	¶	A	·kind	·man	·saw	·L	ily	·cry	ing	·and	·came		
·to	·help	.	·He	·talked	·to	·the	·dog	·and	·said	,	·"	Please	·be	·kind	·to		
·L	ily	."	·The	·dog	·stopped	·b	ark	ing	·and	·did	·not	·b	ite	·L	ily	.	

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
168	113.2	1.48	42	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories15M (Llama-2 architecture) · dim 288 · 6 layers · 6 query / 6 KV heads · 32,000-token vocab
STOPPED BY	End-of-story token, after 168 tokens

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.